

Xinru (Raya) Cheng

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EDUCATION

Peking University

Beijing, China

Master of Software Engineering

Sep 2024 – Jul 2027

- Major Courses: Algorithm analysis and design, Game engines and game development, Natural language processing, Machine learning, Artificial intelligence practices, Software implementation technology, etc.
- Rank: 24/279 (Top 10%)

Beijing University of Chemical Technology

Beijing, China

Bachelor of Digital Media Art

Sep 2020 – Jul 2024

- Major Courses: Programming Foundation, Advanced Mathematics, Game Planning and Development, Advanced Practicum, Virtual Reality Technology, Three Dimensional Animation Design and Production, Graphic Image Processing, Practicum in Applied Software, Engineering Graphics, etc.
- Rank: 1/30 (Direct Admission to Peking University)

ACADEMIC PAPERS

Full Papers

- **Cheng, X.**, Zhang, Y., Wei, Y., Jiang, R., Hansen, P., Liang, H. N. Effects of Spatial Constraint on Bare-Hand Logographic Character Handwriting Performance in Virtual Reality. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J] (Accepted).
- Wei, Y., **Cheng, X.**, Shi, R., Liang, H. N. Reorienting with the Bare Hand: Gesture-Based Techniques for Orientation-Enabled Teleportation in Virtual Reality. In *IEEE Conference on Virtual Reality and 3D User Interfaces (VR'26)* [C] & *IEEE Transactions on Visualization and Computer Graphics (TVCG)* [J].
- Gao, Y., **Cheng, X.**, Li, Y., Xu, D., Sun, D., Hu, Y. ISDG-Net: Efficient RGB-Infrared Object Detection for Remote Sensing Imagery. In *Remote Sensing* [J].
- Zhang, Y., **Cheng, X.**, Yan, J. AI-Echo: Exploring Technological Alienation Through an Interactive Art Experience. In *International Conference on Human-Computer Interaction (HCI'25)* [C].
- Jiang, R., Zhang, S., Zhao, Y., **Cheng, X.**. The Application of Serious Games in Inclusive Education. In *Game Industry Blue Book: China Game Industry Development Report (2025)*. Beijing: Social Sciences Academic Press [BC].

Short Papers

- **Cheng, X.**, Zhang, Y., Qiao, Y., Hansen, P., Jiang, R. Negative Danmaku as Social Warning Cues in Video Credibility Assessment. In *European Conference on Computer-Supported Cooperative Work (ECSCW'26)* (Poster) [C].
- Zhang, Z., Zhang, Y., **Cheng, X.***. Private Non-Visual Addressivity Cues for Blind and Low Vision Users in Mixed-Ability Social VR. In *Conference on Computer-Supported Cooperative Work (CSCW'26)* (Poster) [C] (Accepted).
- **Cheng, X.**, Wei, Y., Liang, H. N. Toward AI-Mediated Candidate Delivery for Chinese Pinyin Input in Virtual Reality. In *1st International Workshop on AI-Mediated Heads-Up Interaction in Wearable Mixed Reality (AHIMR'26)* & In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR'26)* (Workshop) [C] (Accepted).
- Li, Z., Huang, K., **Cheng, X.**, Tang, H., Li, H., Zhao, S., An, P. FlowWalk: AR Guided Outdoor Mindful Walking with Photo Capture Based Nature Awareness. In *ACM International Joint Conference on Pervasive and Ubiquitous Computing / International Symposium on Wearable Computers (UbiComp/ISWC'26)* (Poster) [C] (Accepted).

- Li, Z., Huang, K., Tang, H., **Cheng, X.**, Li, H., Zhao, S., An, P. Demonstrating FlowWalk: A Smart Glasses System for Outdoor Mindful Walking and Connection with Nature. In *ACM International Joint Conference on Pervasive and Ubiquitous Computing / International Symposium on Wearable Computers (UbiComp/ISWC'26)* (Demonstration) [C] (Accepted).

* Corresponding author.

PATENT

- Zhang, P., Li, W., Tang, Z., Wang, Z., Jin, F., **Cheng, X.** *An AI Agent System and Collaborative Knowledge Construction Method*. Chinese Invention Patent, ZL 2025 1 1737406.2, Granted May 19, 2026.

RESEARCH EXPERIENCE

Research Assistant, Computational Media and Arts Thrust, Hong Kong University of Science and Technology (Guangzhou) Jul 2025 – Present

Supervisor: Hai-Ning Liang

- Conducted experimental design, system development, and user studies across three VR projects investigating spatial resolution, gesture-based reorientation, and bare-hand handwriting interaction.
- Contributed to seven manuscripts, including three first-author works, with accepted or published papers in IEEE TVCG and at IEEE VR and IEEE ISMAR.

Research Intern, Department of Computer Science and Engineering, Hong Kong University of Science and Technology Dec 2025 – Present

Supervisor: Xiaojuan Ma

- Co-conceptualized an AI-supported intent-aware writing system, iteratively refined LLM prompts, and designed and conducted a user study to evaluate the interaction.
- Designed and developed TowerBloxx, an iterated Prisoner's Dilemma experiment on human-AI cooperation; implemented the interface, backend, data logging, and dashboard, and conducted a Prolific study with 300+ participants.

Visiting Student, Department of Information Management, Peking University Feb 2025 – Jun 2025

Supervisors: Pengyi Zhang and Preben Hansen

- Led the interaction design and front-end development of two AI-agent-based educational systems supporting collaborative lesson planning, interactive learning, and knowledge construction.
- Contributed to an NSFC-funded research project and a granted Chinese invention patent on AI-supported collaborative knowledge construction.

Research Intern, National Virtual Simulation Experimental Teaching Center for Earth Sciences, Peking University Dec 2024 - Jun 2025

Supervisor: Yanjun Guo

- Conceptualized the study, formulated research questions, and synthesized related work on presence, connectedness to nature, and pro-environmental behavioral intentions in immersive marine science education.
- Recruited 90 participants, conducted the immersive VR user study, managed data collection, and performed statistical analyses.

Research Intern, Future Lab - Space and Media Group, Tsinghua University May 2023 – Jul 2023

Supervisor: Keyang Tang

- Contributed to the interaction design and prototype development of Digital Luoyang, exploring automated generation of architectural and landscape visualizations and creating spatial and animation designs for the virtual environment.

INDUSTRIAL EXPERIENCE

Investment Intern, Alpha Startups

Aug 2025 – Nov 2025

- Conducted pre-investment research, focusing on next-generation interaction technologies and products.

- Participated in all stages of the investment process, including company analysis and market research.
Research Intern, MiraclePlus (YC China) Oct 2024 – Nov 2024
- Responsible for communicating with entrepreneurs and planning Demo Day events.
Product Manager Intern, Kuaishou Technology Apr 2024 – Aug 2024
- Worked on product development for the e-commerce C-end product department.
Product Manager Intern, DiDi Global Nov 2023 – Apr 2024
- Worked on growth product initiatives for the Didi platform, Chitu, and Didi Alliance.
Interaction Design Intern, Meituan Jul 2023 – Nov 2023
- Responsible for interaction design and AI capability building in Meituan Cuisine.
- Received a full-time offer after internship.

ACADEMIC SERVICE & TEACHING

- Teaching Assistant, Game Development and Game Engines, Peking University Sep 2024 - Jan 2025
- Teaching Assistant, Artificial Intelligence Practice, Peking University Mar 2025 - Jul 2025
- Reviewer: CHI 2026, UIST 2026, ISMAR 2025-2026, CHI PLAY 2026, and C&C 2026

HONORS & AWARDS

- **National Scholarship (top 1%)**, 2022.
- Shenzhen Stock Exchange Scholarship, 2025 (one of 17 recipients university-wide).
- Outstanding Student Award, Peking University, 2025.
- People's Scholarship (top 5%), 2021,2022,2023.
- Best Student Short Film, Hollywood Independent Film Festival, 2022.
- Third Prize, National College Student Advertising Art Competition, 2022.

SKILLS

- **Programming & Development:** Python, C#, C++, Blueprint, JavaScript, HTML, CSS
- **XR & Creative Technologies:** Unity, Unreal Engine, Processing, TouchDesigner, Kinect, Leap Motion, Arduino
- **3D Modeling & Rendering:** Blender, 3DS MAX, Substance Painter, Zbrush, Octane Render, Marmoset Toolbag, PBR Material Workflow
- **Design & Media Software:** Figma, Adobe Photoshop, After Effects, Animate, Premiere Pro, Illustrator
- **Statistical Analysis:** Descriptive Statistics, Hypothesis Testing (t-test, Chi-Square Test, ANOVA), Regression Analysis, Non-Parametric Tests
- **Data Analysis:** SPSS, Python (pandas, statsmodels), Excel
- **Hobbies:** Pilates, Diving, FPS Games, Piano(Grade 10 of Wuhan Conservatory of Music)